



Department of Computer Science and Engineering

B.Tech. Computer Science and Engineering

Course Name : Advanced Computer Architecture

Course Code : TCS704

Semester : VII

Pre-requisite : Computer Organization (TCS-404)

L	T	P	Credits
3	0	0	3

Course Learning Objectives:

- Develop an understanding of computer architecture fundamentals, technology trends, and performance evaluation techniques.
- Acquire knowledge of memory hierarchy design, cache organization, virtual memory, and memory optimization techniques.
- Understand pipelining concepts, RISC processor architecture, and pipeline performance issues in modern processors.
- Develop an understanding of branch prediction techniques, instruction-level parallelism, and processor optimization methods.
- Acquire knowledge of multiprocessor architectures and parallel processing models for high-performance computing systems.

Course Outcomes:

After completion of the course, the students will be able to:

Course Outcome	Statement	Bloom's Level	Skill
CO1	Explain the fundamentals of computer architecture, technology trends, performance metrics, benchmarking techniques, and performance evaluation laws.	L2 – Understand	Conceptual Understanding, Analytical Thinking
CO2	Analyse memory hierarchy, cache organization, cache optimization techniques, virtual memory, and address translation mechanisms for improving system performance.	L4 – Analyze	Memory Analysis, Performance Optimization
CO3	Analyse pipelining concepts, RISC pipeline architecture, pipeline performance, and pipeline hazards in modern processors.	L4 – Analyze	Processor Analysis, Problem Solving



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CO4	Analyse branch prediction techniques, instruction-level parallelism, data dependencies, and register optimization methods for enhancing processor performance.	L4 – Analyze	Critical Analysis, Performance Optimization
CO5	Evaluate multiprocessor architectures, shared-memory and distributed-memory systems, and parallel processing models for high-performance computing applications.	L5 – Evaluate	System Evaluation, Parallel Computing Analysis

Details of the Course:

Unit 1: Fundamentals of Computer Architecture and Performance Evaluation

Contact Hours/Teaching Hours: 10

Fundamentals: Computer Architecture and Technology Trends, Moore's Law, Classes of Parallelism and Parallel Architectures, Instruction Set Architecture: The Myopic View of Computer Architecture, Trends in Technology, Trends in Cost, Processor Speed, Cost, Power, Power Consumption, Fabrication Yield Performance Metrics and Evaluation: Measuring Performance, Benchmark Standards, Iron Law of Performance, Amdahl's Law, Lhadma's Law.

Self-Learning Hours = 7

Students should study the fundamentals of computer architecture, understand the evolution of processor technology, analyze Moore's Law and technology trends, compare different parallel architectures, explore instruction set architectures, learn the impact of cost, power, and fabrication yield on processor design, and evaluate system performance using benchmark standards, the Iron Law of Performance, Amdahl's Law, and Lhadma's Law.

Team Activity Hours = 3

Students should work in teams to compare different computer architectures and parallel processing models, analyze benchmark results of modern processors, demonstrate the impact of performance metrics on system efficiency, and present case studies illustrating the application of Amdahl's Law and performance evaluation techniques.

Unit 2: Memory Hierarchy Design

Contact Hours/Teaching Hours: 9

Memory Hierarchy Design: Basics of Memory Hierarchy, Coherence and Locality Properties, Cache Memory Organizations, Cache Performance, Cache Optimization Techniques, Virtual Memory, Techniques for Fast Address Translation.

Self-Learning Hours = 6

Students should study the organization of the memory hierarchy, understand locality and coherence properties, compare different cache organizations, analyze cache performance and optimization



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techniques, explore virtual memory concepts, and learn fast address translation mechanisms for improving memory access efficiency.

Team Activity Hours = 3

Students should work in teams to analyze different cache organizations, demonstrate cache mapping techniques using examples, evaluate cache optimization methods, and present the role of virtual memory and address translation in enhancing system performance.

Unit 3: Pipelining

Contact Hours/Teaching Hours: 10

Pipelining: What is Pipelining, Basics of a RISC ISA, The Classic Five-Stage Pipeline for a RISC Processor, Performance Issues in Pipelining, Pipeline Hazards.

Self-Learning Hours = 7

Students should study the concept of instruction pipelining, understand the RISC instruction set architecture, learn the operation of the five-stage pipeline, analyze pipeline hazards and performance issues, and explore techniques for improving instruction execution efficiency.

Team Activity Hours = 3

Students should work in teams to demonstrate instruction execution through a pipelined processor, identify different pipeline hazards, apply hazard resolution techniques, and present the performance improvements achieved through pipelining.

Unit 4: Branch Prediction and Instruction Level Parallelism

Contact Hours/Teaching Hours: 8

Branches and Prediction: Branch Prediction, Direction Predictor, Hierarchical Predictors, If Conversion, Conditional Move. Instruction Level Parallelism: Introduction, RAW and WAW Dependencies, Duplicating Register Values, ILP.

Self-Learning Hours = 6

Students should study branch prediction techniques, understand direction and hierarchical predictors, explore conditional execution methods, analyze instruction dependencies, learn instruction-level parallelism concepts, and evaluate techniques for improving processor performance.

Team Activity Hours = 3

Students should work in teams to compare different branch prediction techniques, identify instruction dependencies in sample programs, demonstrate instruction-level parallelism using scheduling examples, and present methods for improving processor throughput.



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Unit 5: Multiprocessor Architecture

Contact Hours/Teaching Hours: 9

Multiprocessor Architecture: Taxonomy of Parallel Architectures, Centralized Shared-Memory Architecture, Distributed Shared-Memory Architecture, Message Passing vs Shared Memory.

Self-Learning Hours = 6

Students should study the taxonomy of parallel architectures, understand centralized and distributed shared-memory systems, compare message-passing and shared-memory communication models, and evaluate their advantages, limitations, and applications in parallel computing environments.

Team Activity Hours = 3

Students should work in teams to compare different multiprocessor architectures, analyze the advantages and limitations of shared-memory and message-passing systems, prepare case studies on parallel computing applications, and present suitable architecture choices for different computing scenarios.

Course Articulation Matrix

Programme Outcomes Course Outcome	PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PO 12	PS O01	PS O02	PS O03
CO01	3	2	1	1	2					1		1	2	3	1
CO02	3	3	2	2	3							1	3	3	2
CO03	3	3	2	2	3							1	3	3	2
CO04	3	3	2	2	3					1		1	3	3	2
CO05	3	3	3	2	3			1		1		2	3	3	3
Avg	3	3	2	2	3			1		1		1	3	3	2

Text Books:

1. John L. Hennessy, David A. Patterson, "Computer Architecture: A Quantitative Approach" 5 th edition, Morgan Kaufmann "by Kai Hwang , "Advanced Computer Architecture", McGraw Hill Publishing

Online Resources:

<https://www.geeksforgeeks.org/computerarchitecture>